rubber

Frank Parmenter winding his Charisma Wakefield. He is using a protective tube in the fuselage and an extension onwinder, permitting the removal of the tube. Protection is a must on big rubber models, because a broken motor while winding will have catastrophic results.

IN A previous article the development of several new types of excellent quality rubber was described. Laboratory tests and field tests with models have demonstrated that these new rubber types are equivalent in quality to pre-1973 Pirelli. Since 1973, when the Pirelli company reorganized and disrupted the supply of rubber used by most contestants, there was no consistent supply of good rubber strip for model airplanes until 1976. In that year FAI rubber appeared and, a little later, Sig and Vintage Aero rubber with similar properties. In 1978 Pirelli arrived back on the scene with the new Pirelli especially made for model fliers. The new Pirelli is yellow-brown and almost translucent, whereas, the Filati is dark brown and opaque. The excellent quality of these new types of rubber according to Mike Dikovitzky has, "Put the fun back into flying rubber powered models!"

Table I gives the characteristics of a typical batch of FAI rubber. Table II gives the characteristics of a batch of the 1978 new Pirelli. The FAI rubber is thicker than the Pirelli. Typically, 14 Strands of FAI is equivalent to 16 Strands of Pirelli. The stretchiness of rubber will vary considerably from batch to batch, and is closely correlated with the capability to take turns. Good rubber varies from a maximum stretch in terms of original length of 7.3 to 8.1. Be suspicious of rubber with more or less stretchiness than these figures.

If it is less than 7.3 it is probably old and can be brittle. If it is above 8.2 it can be undervulcanized or too soft. Undervulcanized rubber has a torque curve which drops off more rapidly than good rubber. This gives trimming problems and leads to mushing and stalls shortly after launch, although the power towards the end of the motor run holds up very well.

Expect to find differences from batch to batch of both good types of rubber and inferior types of rubber. Tests of Pirelli from 1958 t o 1972 ranged in stretch from 7.4 to 8.4 and energy ranged from 2940 to 3525. Current tests of FAI rubber indicate as wide a range of properties as encountered with Pirelli and with the average energy as good as the pre-1973 Pirelli. At the time of this writing, experience with the new Pirelli was limited to one batch. It was superior in energy to all previously tested rubber, giving 3725 foot-pounds per pound. This rubber gave a significant increase in duration as was demonstrated in the down 4-minute rounds at the 1978 Taft Wakefield Eliminations. Although the new Pirelli was the obvious choice for cool early morning flying, the few fliers, with supplies of this scarce new product, experienced difficulty later in the day. The Pirelli tended to blow up when the temperatures got over 90°F. This led those with both Pirelli and FAI to switch to FAI rubber, which is much less likely to blow-up. This leads to another topic. What effect does ambient temperature have on the ability of rubber to store energy?

Ambient Temperature Effect: The higher the ambient temperature, the more energy can be stored. In addition, the shape of the power curve is better at higher temperature.

Fig. 6 shows the change in the power curve. There is more energy in the knee of the power curve. With more energy in the knee, the airplane will perform better through the period a few seconds after launch when stalls are more likely. At present the writer has a limited amount of temperature effect data. Fig. 7 presents the picture. Both FAI and Pirelli have about 5% more energy at 90°F than at 75°F. Below 75°F the characteristics diverge. The Pirelli rate of change remains about the same as at higher temperature, but the FAI declines more rapidly, being down perhaps 20% at 40°F. It is thought that these figures vary from batch to batch. (Walter Erbach discussed temperature effect in his excellent report "Energy Storage in Rubber Filaments" which was published in the 1971 NFFS Symposium.)

Since the rubber stores more energy when it gets hotter, why not try to keep rubber as hot as possible? When being

-- All You Need to Know

Last month the author reported on his testing program—and its findings. Concluding two-part coverage, he discusses handling for optimum results. ■ Fred Pearce

wound in the sun, the exposed rubber gets significantly hotter than the ambient temperature. Experience indicates that above 95°F, when winding in the sun, that you are much more likely to break a motor. So, be careful when trying to take advantage of rubber's capability to store more energy when it is warmer. In cold weather it is much more of an advantage to use Pirelli than FAI.

Maximum Turns: Tables I and II give the maximum turns for several skein sizes for FAI and the new Pirelli of specific batch. The numbers derive from actual winding experience with these batches. Different batches, of course, will vary in the number of maximum turns.

The formula for maximum turns is:

$$\text{Turns (max)} = K\left[\text{Original Length, in.}\right]\left[\frac{\text{Original Length}}{\text{Weight, oz.}}\right]^{1/2}$$

the constant K varies as the stretchiness of the rubber, and can be approximated by the formula:

$$K = .75\left[\frac{\text{Maximum Stretched Length}}{\text{Original Length}}\right]$$

If you know the maximum turns for a given size motor from a specific batch, you can solve for K in the first formula and then compute turns for any size motor from that batch. The K for the two batches in the tables is given at the bottom of each table.

TABLE I
Motor Characteristics of March 1977 FAI Rubber *

Type	Dry Weight	Strands	Original Length	Broken-in Length	Max Turns
Wakefield	38.8 gm.	10 x ¼"	22.8"	23.9"	530
"	"	12 x ¼"	19.0"	19.9"	403
"	"	14 x ¼"	16.3"	17.1"	320
"	"	16 x ¼"	14.3"	15.0"	263
Coupe d'Hiver	9.7 gm.	10 x 1/8"	11.4"	12.0"	375
"	"	6 x ¼"	9.51"	9.98"	286

* Energy was 3450 foot pounds per pound and the maximum stretch was 7.6 times the original length. Turns constant K = 5.70.

TABLE II
Motor Characteristics of 1978 New Pirelli Rubber *

Type	Dry Weight	Strands	Original Length	Broken-in Length	Max Turns
Wakefield	38.8 gm.	10 x ¼"	26.0"	27.0"	686
"	"	12 x ¼"	21.7"	22.6"	523
"	"	14 x ¼"	18.6"	19.3"	415
"	"	16 x ¼"	16.3"	17.0"	341
Coupe d'Hiver	9.7 gm.	10 x 1/8"	12.9"	13.4"	480
"	"	6 x ¼"	10.75"	11.2"	365
"	"	14 x 1/8"	9.27"	9.63"	293

*. Energy was 3725 foot pounds per pound and the maximum stretch was 8.1 times original length. Turns constant K = 6.06.

Rubber-holding stirrups for enabling detachment of prop assembly for winding. Top one is Wakefield, the bottom one for Coupe.

Strand Size: Does the width of the rubber affect the energy storage? Would you get more or less energy by using an equivalent skein of many smaller strands? It has been hypothesized that a larger number of smaller strands to the same cross-section area would enable one to store more energy, since the tensile stress across the narrow strands would be more even, or have less gradient. This question was intriguing enough to induce Bob Champine to conduct a test on Pirelli about 15 years ago. He divided a sample of ¼" rubber into two halves and stripped one half into 1/10" wide strand. Both motors were then made up to equivalent cross-section. Energy storage tests were performed using the torsion method. The unwinding torques vs. turns were recorded. No appreciable difference in torque or energy storage was evidenced. Thus, the only trade-off in using different widths of rubber, other than to get the proper cross-section, is that smaller sizes tie easier knots, but they also offer more edges in a skein to get frayed.

Tying-up the Skein: Table I will be found useful in sizing the skeins desired. Special care must be taken in tying the knot because of the smooth surface of the strip. Wetting the ends and dusting them with extra talcum powder helps make a secure knot which will not slip. A square knot topped by a single granny works well. Saliva generously applied aids in tying the knots.

Breaking-in: The lubricated motors should be broken in by stretching prior to use. A force of 7 pounds of pull per strand of ¼" is suggested (98 pounds for a 14-strand motor). This amount of pull will stretch the motor between 7 and 8 times its original length in

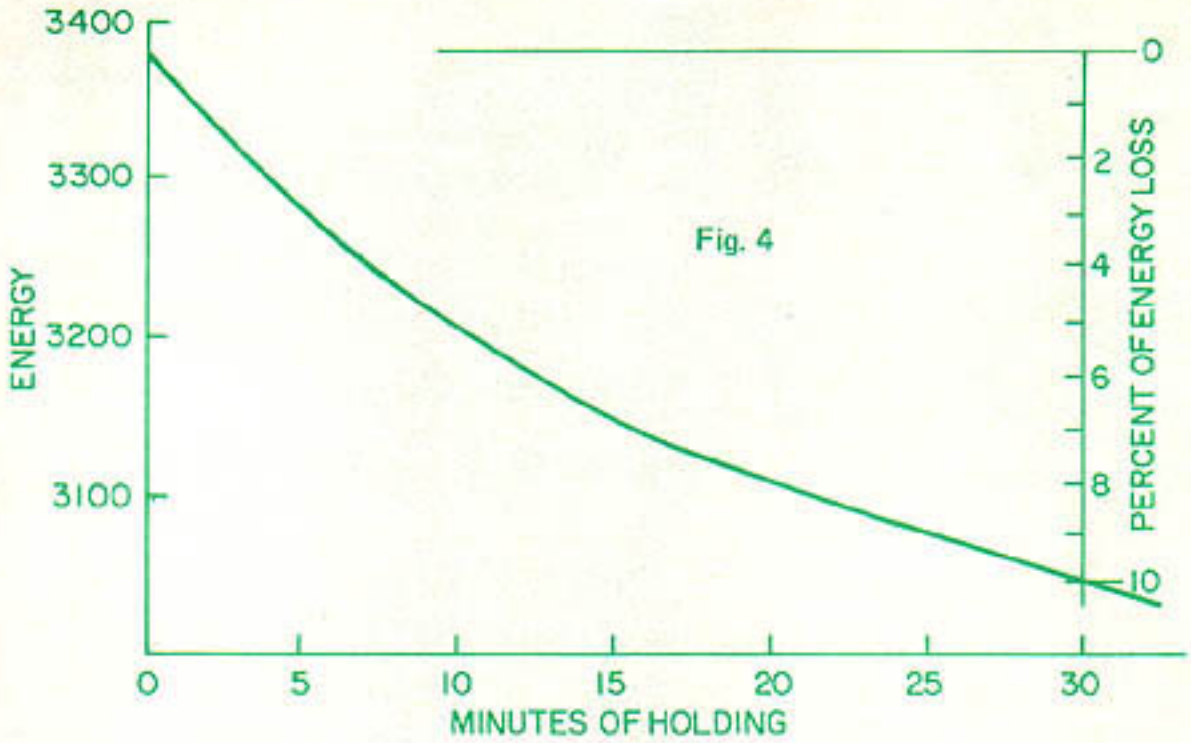
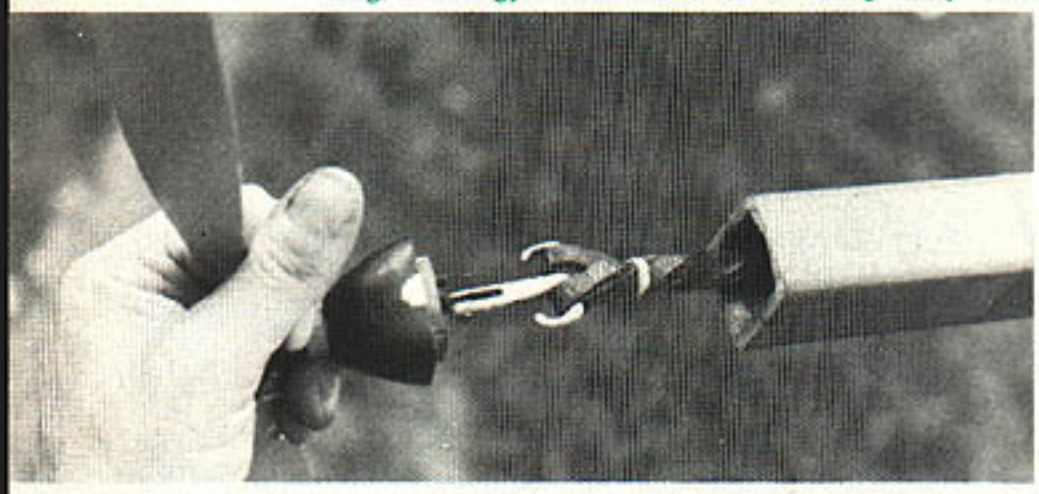


Fig. 4: Energy loss with time when holding a fully wound motor.

several minutes. I pull the motors twice. Once very quickly to 7 lb/strand relaxing immediately; then a second time, holding the motor for 5 minutes. It is important to have a secure anchor point. A second anchor point for the end you pull is equally important. Inspect and re-tie any slipped or torn knots. If measuring the pull force is impractical for you, stretch to at least 7 times the original length and hold for 5 minutes.

Lubrication: A green soap and glycerine mixture is recommended. The green soap is hard to find, although it can occasionally be found in one-pound jars. It resembles brown grease in this form. More often, only tincture of green soap is available. All the alcohol must be boiled out of the tincture as it is very corrosive to rubber.

To prepare the lubricant, boil 8 ounces of tincture in a sauce pan at low heat. After it boils for about 20 minutes, add about 3 ounces of glycerine and boil another 20 minutes at low heat. Adjust the viscosity to that of thin honey by adding glycerine after



The front end pulled out to show the way the stirrup holds the rubber.

Propeller assembly showing aluminum stirrup on prop shaft. The stirrup enables the propeller assembly to be detached for safety in winding.

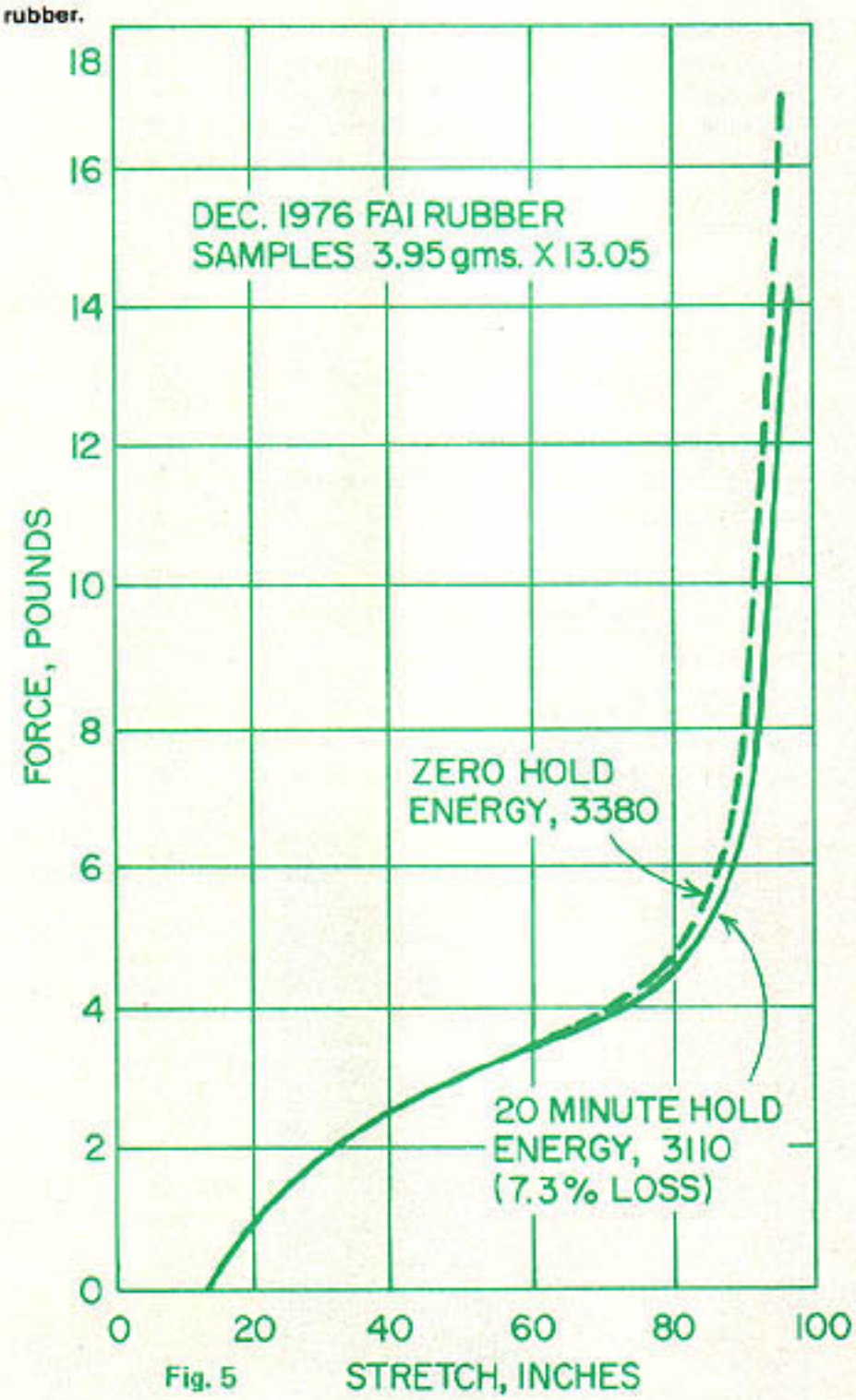


Fig. 5

Fig. 5: Comparison of energy before and after holding wound up motor 20 min.

it cools.

Be careful in getting the lubricant evenly distributed. Lubricate the motors before breaking them in. Smooth rubber is inclined to squeeze itself dry when wound; to prevent breakage redistribute the lubricant evenly after each winding. (Table I allows for 1.2 grams of lubricant to bring the rubber up to weight for the Wakefield-size motor.)

Winding: A sturdy winder with good bearings is helpful in winding. The author converted a large hand-drill to a winder by installing a thrust bearing to take the pull of the motor, and by adding additional leverage to the crank. (See photograph.) Several supply sources sell winders.

Begin by stretching the motor to about 4 times its length. At this point, wind in 50 to 55% of the maximum turns. Continue winding slowly all the way in for the full count. Bunching around the front hook or bobbin is aggravated by stopping winding several inches from the nose and coming the rest of the way without winding.

The author uses a special bobbin and tube assembly as shown in the photographs. This array permits winding with the propeller removed and with an aluminum protective tube in the fuselage to prevent damage by broken motors. Remember the caution about redistributing the lubricant between windings. If you don't redistribute, tearing will take place at the dry spots.

Holding a Wound Motor: An important quality factor of rubber strip is its ability to maintain its energy storage while being held fully wound. In a contest, it is frequently necessary to wait many minutes for a thermal. Rubber which does not decline rapidly in stored energy is a great asset under these conditions. The FAI rubber is exceptionally good in its capability to hold energy. Figure 4 shows the decline in stored energy of this rubber with holding time. It loses power in a rather linear manner with a 5% loss after 11 minutes and a 10% loss at 30 minutes. Jon Davis in his 1971 NFFS Symposium article reported on Pirelli, which he found declined much more rapidly than FAI rubber. His Pirelli tests indicated a 15% loss in six minutes. Figure 5 shows the change in the power curve when holding a fully wound FAI rubber motor for 20 minutes. The rubber lost 7.3% of its energy, chiefly at the "knee" of the curve—corresponding to the first 5 seconds of a motor run.

It is highly probable that the capability to hold energy varies significantly from batch to batch and with ambient temperature. Comments at Taft on the holding quality of the new Pirelli were that it was not as good as the FAI at holding its energy.

A Few Closing Comments: The author found that it was extremely beneficial to have a winder with a thrust to take the pull force. With a low-friction winder, you can feel the torque of the motor and can maintain

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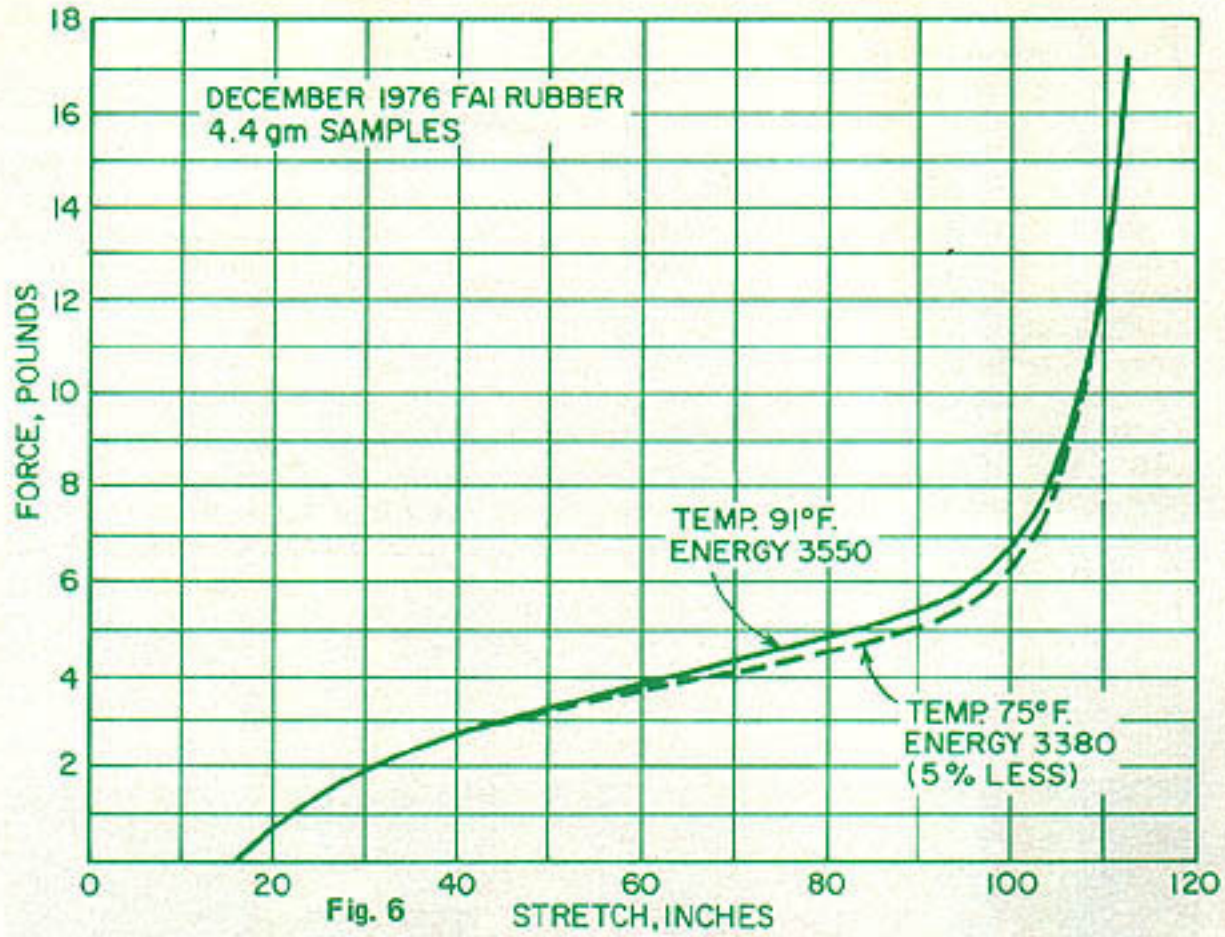
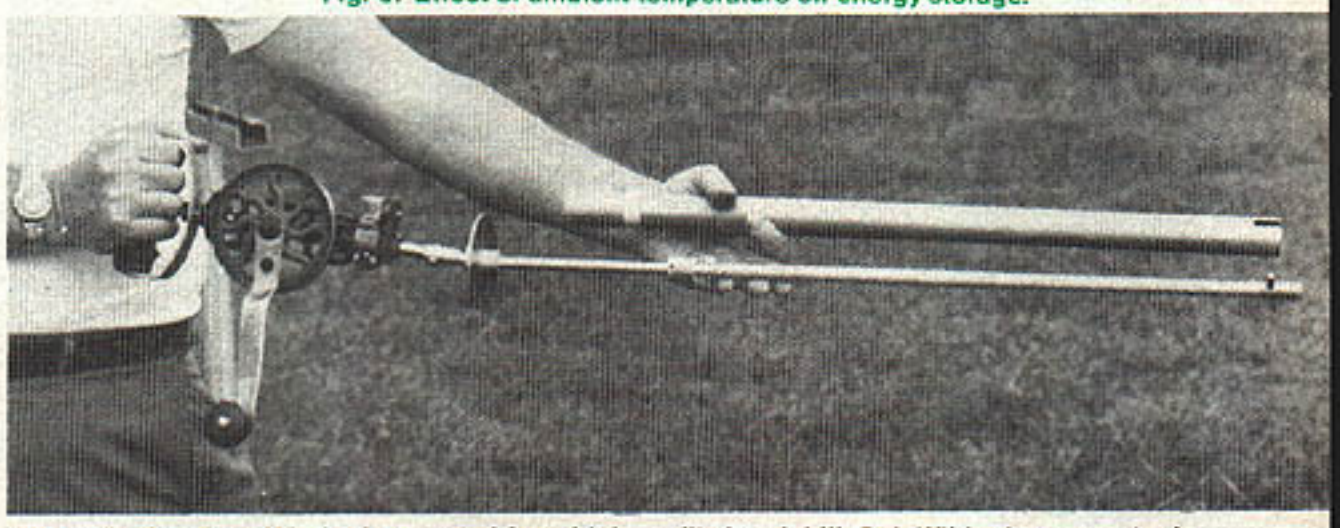
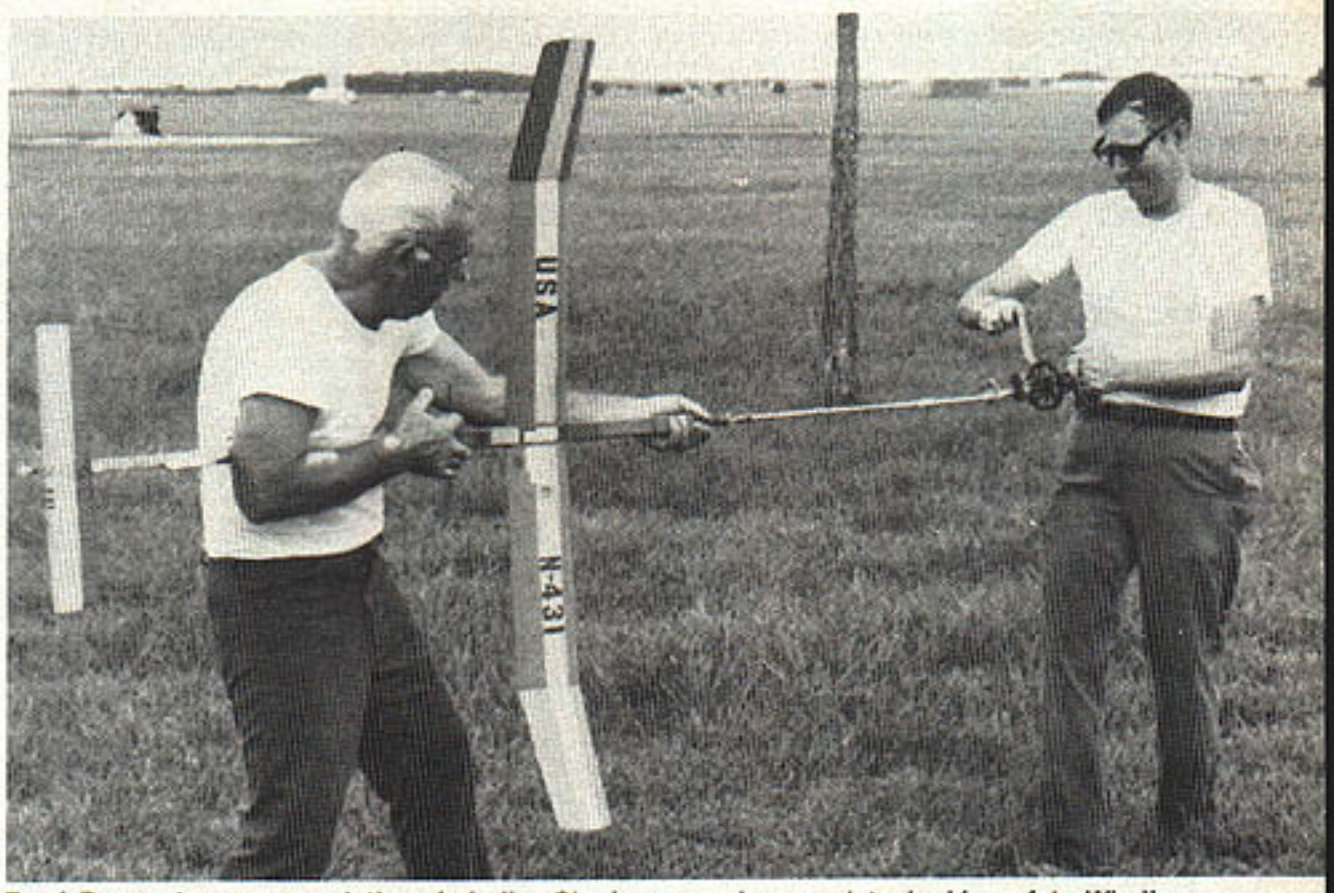


Fig. 6: Effect of ambient temperature on energy storage.



Motor winding gear. Winder is coverted from high quality hand drill. Bob Wilder torque meter is atached to winder, as well as an aluminum rod. The rod enables removal of protective aluminum tube from inside the fuselage after winding—with propeller detached for safety.



Frank Parmenter near completion of winding Charisma—again, prop detached for safety. Winding began with motor stretched out the nose four motor lengths. Bill Shaw holds with aid of ⅛ in. steel rod, through ¼ in. aluminum motor anchor. Long rod attached to winder enables removal of 1¼ in. protective aluminum tube from inside the fuselage after the motor has been wound.

subjects with the landing gear length as a prime factor? This is ridiculous. Are we building "flying scale," or "takeoff scale," models?"

An excellent method of hand-launching is proposed by Ken Hamilton, who has noticed many otherwise accomplished scale modelers launching with excessive speed and/or nose-up angles and with the hand moving through an arc. The model, says Ken, should be projected "... along a straight line (the desired flight path) by means of an arm motion involving both the shoulder and elbow joints as the arm moves the hand in a straight line." If one can match the flying speed expected from the model, excess lift (produced by nose-up hurling) will be minimized. Sort of just let it fly out of your hand. Ken, one of the top men in the aircraft industry (Sabreliner project) before his retirement, adds: "Real aircraft are docile within a design speed envelope, but become unruly if pushed outside it, too!"

This month's TWA comes from Dick Seifried. His Curtiss XP-37 (pic in Jan. MA) was lost at the 1978 Flightmasters' Annual with three flights of about a minute each and a final awe-inspiring 14 minutes OOS! At 22" span, the one-ounce model was built "a la Earl Stahl" with clamshell construction. It flew on six strands of ⅛ × 22" Sig contest rubber with one-minute motor runs. The model did best in calm air, but had trouble with ground turbulence due to the low power. Dick feels that a shorter or more powerful motor would get it up and out of this problem, but with a reduction of motor run to 35/40 seconds. Dick offers to loan the photo negative for making your own plans! His address is R. Seifried Jr., 13022 Barto Dr., Granada Hills, CA 91344.

Many readers have sent in recommendations that the best buy of the year is a gold mine of rubber-scale plans reduced in a catalogue by Golden Age Reproductions, P.O.Box #13, Braintree, MA 02184. Put out by Joe Fitzgibbon, the book shows plans which you can order from him full size, but the collection of the 150 vintage drawings is more than worth the price of the catalogue! Send along $5. You won't regret it!

Keep the great hints, tips, TWA's and 5 × 7 B/W's coming!

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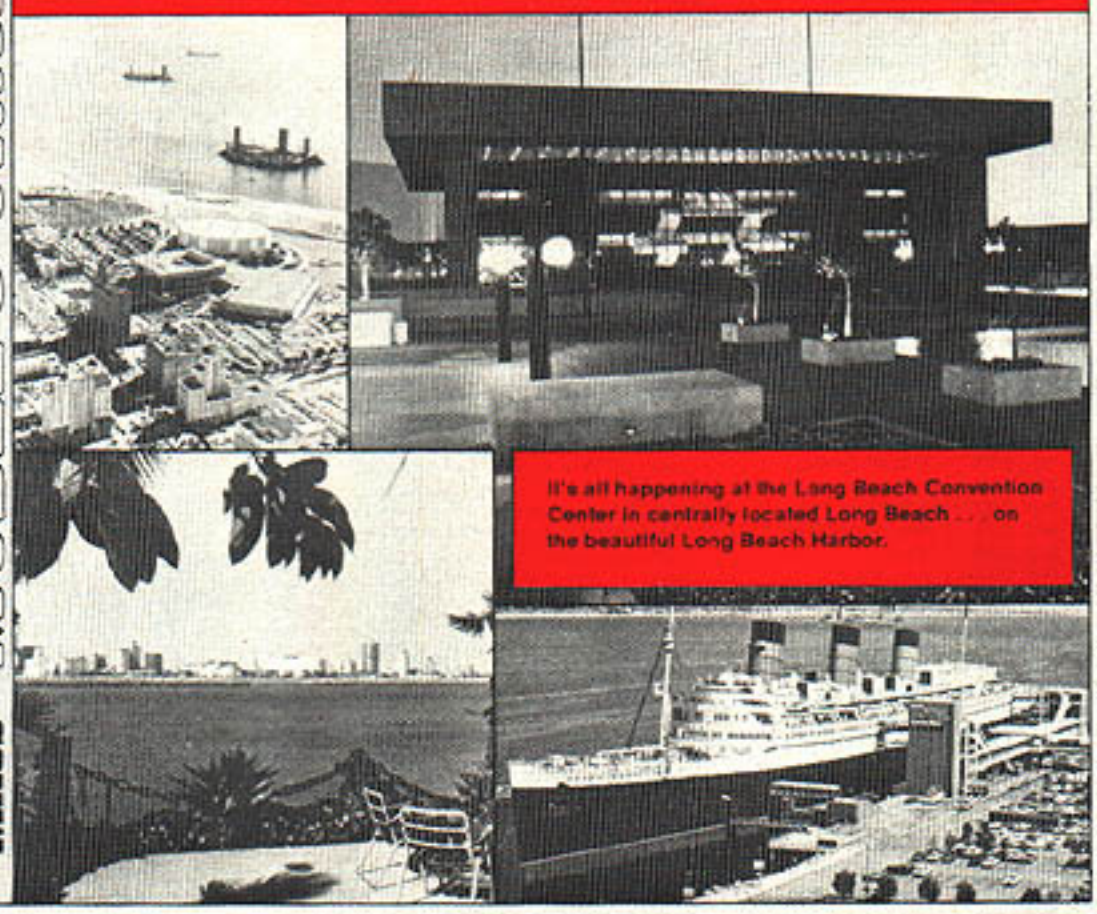

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Rubber/Pearce

continued from page 71

closer control when winding to maximum turns. It is a great priviledge to wind knowing that, if you blow-up the motor, your model fuselage will survive. The author achieved this with a protective tube system as shown in the photographs. Others have constructed fuselages of layers of balsa and fiberglass which can contain the broken motor.

Experience with the new Pirelli is limited to small quantities of a single batch. A source in the U.S. for the Pirelli was not established when this was written in September 1978. It is not a perfect product. It has bad spots and edge defects, as does FAI rubber. The choice between Pirelli and FAI is not a simple choice. Both have their virtues. Although the Pirelli has about 9% more energy at 75°F, the FAI retains its energy better when you hold while waiting for good air. If you hold over five minutes, it

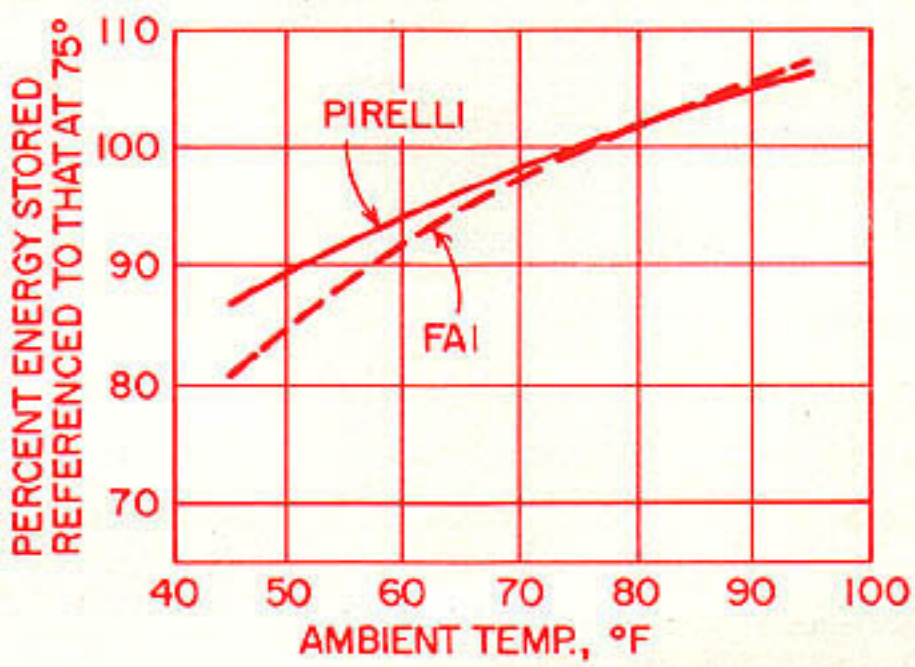

Fig. 7: Energy storage variation with temperature.

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is probable that the FAI will have more energy, since the FAI loses energy at a much lower rate. The Pirelli, despite its 9% higher energy storage, proved treacherous in very hot weather at Taft when frequent breaking was encountered. But, we have experience with only one batch of new Pirelli. Who knows what subsequent batches may be like!

FAI rubber can be obtained from FAI Model Supply, P.O.Box 9778, Phoenix, AZ 85068. The new Pirelli can also be obtained from FAI Model Supply and from Oldtimer Models, P.O. Box 18002, Milwaukee, WI 53218.

If you have any questions or news concerning rubber, please contact the author, Fred Pearce, at 5317 Blythewood St., Houston, TX 77021.

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